DISTRIBUTED ALGORITHMS • GRAPH TRAVERSAL • SPANNING TREES

Synchronous Distributed Breadth-First Search (BFS)

Synthesis based on Arpit Bhayani's Distributed Systems Masterclass • Lynch (Distributed Algorithms)

ALGORITHMIC PREMISE & OVERLAY ARCHITECTURE

Unconstrained message flooding in distributed graphs leads to broadcast storms and exponential message duplication. To establish loop-free routing and optimal aggregation, distributed networks construct a **Breadth-First Search (BFS) Spanning Tree**. Under synchronous rounds, Distributed BFS builds shortest-path trees layer-by-layer in concentric expansion waves, terminating in $\mathcal{O}(D)$ rounds with $\mathcal{O}(|E|)$ messages, unlocking optimal Broadcast and Convergecast primitives.

1. THE DISTRIBUTED BFS CHALLENGE

Unlike centralized BFS where a single FIFO queue processes vertices in $O(|V| + |E|)$ time, distributed systems have no shared memory or global clock. Nodes communicate exclusively with immediate neighbors to discover hop distances and assign parent pointers.

2. SYSTEM MODEL & STATE VARIABLES

Network Assumptions

- **Connected Graph:** $G = (V, E)$ with $|V| = n$ and $|E| = m$.
- **Designated Root r:** Source vertex initiating tree creation.
- **Synchronous Rounds:** Lock-step rounds $1, 2, 3, \dots$.

Local Node State (Node u)

- $ ext{parent}_u$: Pointer to upstream parent node.
- $ ext{children}_u$: Verified set of downstream child nodes.
- $ ext{dist}_u$: Minimum hop distance from root r.

3. LAYERED WAVEFRONT PROPAGATION

The construction proceeds via exploration waves moving outward from root r:

- **Round 0:** Root r sets $ ext{dist}_r \gets 0$ and broadcasts `SEARCH(0)` to all neighbors $\mathcal{N}(r)$.
- **Round k (Layer Discovery):** Unvisited node u receiving `SEARCH` sets $ ext{dist}_u \gets k$. If multiple neighbors invite u simultaneously, u selects one via tie-breaking ($ rg\min ext{UID}$) and sends `CONFIRM` to the chosen parent and `REJECT` to all other contenders.
- **Round $k+1$:** Node u broadcasts `SEARCH(k)` to its unvisited neighbors.

4. CONVERGECAST & GLOBAL TERMINATION

How does the root know when the tree is complete? Distributed BFS uses an inward **Convergecast Wave**:

1. Leaf Detection

A node u discovers it is a leaf when all non-parent neighbors reject its search token. It has zero child nodes.

2. Inward Echo

Each leaf sends a `COMPLETED` token upstream to its parent. Parents aggregate echoes from all children.

3. Root Finalization

When the root receives completion tokens from all direct children, tree construction terminates globally.

5. FORMAL COMPLEXITY & OPTIMALITY PROOFS

Time Complexity: $\mathcal{O}(D)$ Rounds

The outward search wave takes $ ext{depth}(T) \le D$ rounds. The inward convergecast wave takes $ ext{depth}(T) \le D$ rounds. Total rounds: $T_{ ext{total}} \le 2D = \mathcal{O}(D)$. This is time-optimal.

Message Complexity: $\mathcal{O}(|E|)$ Messages

Along each edge $\{u, v\}$, search tokens, parent confirmation, and reject messages are exchanged at most a constant number of times (≤ 4 msgs/edge). Total messages $\le 4|E| = \mathcal{O}(|E|)$.

6. COMMUNICATION PRIMITIVES POWERED BY BFS TREES

Primitive	Direction	Message Flow	Time Complexity	Primary Production Use Case
Broadcast	Root → Leaves	Parent to all children	$ ext{depth}(T)$ rounds	Configuration distribution, topic publish
Convergecast	Leaves → Root	Child to parent with reduction	$ ext{depth}(T)$ rounds	Distributed sum, max, barrier sync
Shortest-Path Route	Node → Root	Hop along parent pointers	$ ext{dist}(u, r)$ hops	Telemetry aggregation, log shipping

7. DISTRIBUTED BFS VS. FLOODING VS. BELLMAN-FORD

Algorithm	Edge Weights	Output Structure	Time Complexity	Message Complexity
Naive Flooding	Unweighted	None (Message broadcast only)	$\mathcal{O}(D)$	$\mathcal{O}(\|E\|)$ per broadcast (Storms)
Synchronous BFS	Unweighted (Hop count)	Directed Spanning Tree	$\mathcal{O}(D)$ rounds	$\mathcal{O}(\|E\|)$ (One-time setup)
Distributed Bellman-Ford	Weighted (Positive/Negative)	Shortest Path DAG	$\mathcal{O}(\|V\|)$ rounds	$\mathcal{O}(\|V\| \cdot \|E\|)$
GHS Algorithm	Weighted (Positive)	Minimum Spanning Tree (MST)	$\mathcal{O}(\|V\| \log \|V\|)$	$\mathcal{O}(\|V\| \log \|V\| + \|E\|)$

8. SYSTEMS ENGINEERING TAKEAWAYS

1. **Structural Reduction of Traffic:** Once a BFS tree is constructed, ongoing cluster-wide broadcast drops from $\mathcal{O}(|E|)$ messages down to exactly $n-1$ messages, eliminating broadcast storm vulnerability.
2. **Parent Tie-Breaking Invariants:** In distributed environments, deterministic arbitration (e.g. smallest sender UID) ensures that multiple simultaneous search invitations never induce cycles.
3. **Foundation for MapReduce / Stream Processing:** The split between Broadcast (Map / scatter) and Convergecast (Reduce / gather) forms the core mathematical foundation of distributed data processing engines.